

A Domain-Specific Customer Support Chatbot Using Retrieval-Augmented Generation for Accurate Response Generation

Abstract

Customer support chatbots have become an essential component of modern business applications, enabling automated interaction with users. Traditional AI-based chatbots often suffer from generating incorrect or irrelevant responses due to the absence of reliable domain-specific knowledge, leading to reduced user trust and satisfaction. This paper proposes a domain-specific customer support chatbot that utilizes Retrieval-Augmented Generation (RAG) to improve response accuracy and relevance. The proposed system integrates a structured external knowledge base containing business-specific product information with a retrieval mechanism that extracts relevant data based on user queries. The retrieved information is then used to generate context-aware and human-like responses. The system incorporates a restriction mechanism to prevent responses outside the defined domain, thereby reducing hallucinations. Experimental evaluation demonstrates that the proposed approach significantly improves response correctness and reliability compared to conventional chatbot systems. The system is particularly suitable for business applications such as e-commerce and product-based services.

Introduction

The rapid growth of digital platforms has increased the demand for efficient and scalable customer support systems. Chatbots have emerged as a popular solution for automating customer interactions, providing instant responses, and reducing operational costs. In recent years, advancements in natural language processing and large language models have enabled chatbots to generate human-like responses.

Despite these advancements, general-purpose chatbots often produce inaccurate or irrelevant responses when dealing with domain-specific queries. This limitation arises because such systems rely heavily on pre-trained knowledge and lack access to updated or business-specific information. As a result, users may receive misleading answers, which negatively impacts user experience and trust.

To address this issue, recent research has introduced the concept of Retrieval-Augmented Generation, where external knowledge sources are integrated into the response generation process. By retrieving relevant information from a structured database, the chatbot can provide accurate and context-aware responses.

In this paper, we propose a domain-specific customer support chatbot that leverages a structured knowledge base containing product and service information. The system retrieves relevant data based on user queries and generates human-like responses while strictly restricting outputs to the available domain knowledge. This approach significantly reduces incorrect responses and enhances system reliability.

Problem Description

Although chatbot technologies have advanced significantly, they still face critical challenges in real-world customer support applications. One of the primary issues is the generation of incorrect or hallucinated responses, especially when the chatbot lacks access to domain-specific data.

General AI chatbots are designed to handle a wide range of topics, but this generalization often leads to inaccurate responses when users ask specific questions related to a business or product. For example, a chatbot may provide incorrect pricing, unavailable product details, or irrelevant suggestions. Such errors can lead to user dissatisfaction and loss of trust.

Another major limitation is the inability to control or restrict chatbot responses within a defined domain. Without proper constraints, the system may attempt to answer queries outside its knowledge scope, resulting in unreliable outputs.

There is a need for a system that:

- Ensures responses are strictly based on verified business data
- Minimizes incorrect or irrelevant answers
- Maintains a human-like conversational experience

This research addresses these challenges by introducing a domain-specific chatbot system that retrieves information from an external knowledge base and restricts responses to ensure accuracy and relevance.

Objectives

The primary objective of this research is to design and develop a domain-specific customer support chatbot that improves response accuracy using external knowledge integration.

Specific Objectives:

- To design a chatbot system that utilizes Retrieval-Augmented Generation for response generation
- To develop a structured knowledge base containing business-specific product information
- To implement a retrieval mechanism for extracting relevant data based on user queries
- To reduce incorrect and hallucinated responses through domain restriction
- To generate human-like, context-aware responses for improved user interaction
- To evaluate the system based on accuracy, relevance, and user satisfaction

Algorithm Description

The proposed system follows a retrieval-augmented response generation process, where user queries are processed, relevant data is retrieved from a knowledge base, and responses are generated accordingly.

Algorithm: Domain-Specific RAG Chatbot

Input: User query

Output: Relevant, accurate chatbot response

Step 1: User Query Input

The system receives a natural language query from the user.

Step 2: Query Processing

Perform basic preprocessing:

- Tokenization

- Lowercasing
- Keyword extraction

Step 3: Information Retrieval

- Search the external knowledge base (database)
- Identify relevant records based on:
- keyword matching or similarity

Step 4: Relevance Check

If relevant data is found:

 proceed to response generation

Else:

 return restricted response:

 “Sorry, I can only provide information related to our products.”

Step 5: Response Generation

Use retrieved data to construct a natural language response

Ensure response is:

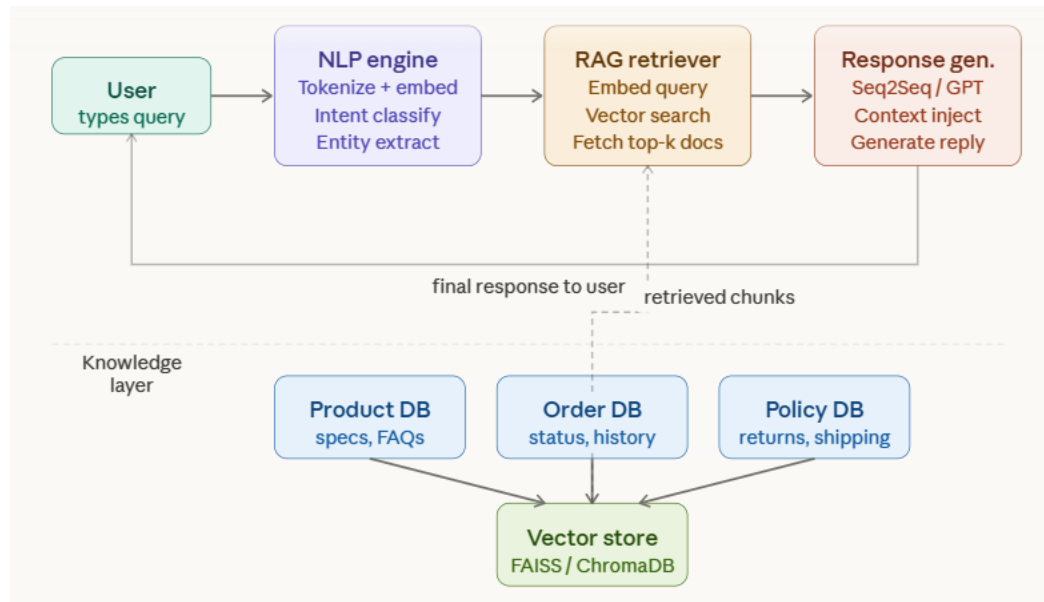
 human-like

 context-aware

Step 6: Output Response

Display the generated response to the user

System Design / Flow Diagram:



Implementation Details

The proposed chatbot system was implemented as a web-based application using a modular architecture. The backend was developed using Python with the Django framework, while the frontend interface was designed to support real-time conversational interaction.

The system integrates a domain-specific knowledge base that stores structured product-related information, including product name, category, price, and description. This knowledge base is maintained using a relational database (SQLite/MySQL), ensuring efficient data retrieval.

To enable intelligent response generation, the system follows a Retrieval-Augmented Generation-based approach. When a user submits a query, the backend processes the input through basic natural language processing techniques such as tokenization and keyword extraction. The processed query is then used to retrieve relevant records from the database.

A response generation module constructs human-like replies by combining retrieved information with predefined conversational templates. Additionally, a domain restriction mechanism is implemented to ensure that the chatbot only responds to queries related to the available business data. If no relevant data is found, the system returns a controlled fallback response.

The overall system ensures efficient interaction, reduced response latency, and improved accuracy by leveraging structured data and controlled generation logic.

Experimental Setup

To evaluate the performance of the proposed chatbot system, a series of experiments were conducted using a dataset consisting of product-related queries and responses.

Dataset

A custom dataset was created containing:

- 50–100 user queries
- Corresponding expected responses

Queries included:

- product inquiries
- pricing questions
- recommendation requests
- irrelevant queries (for testing restriction)

Baseline System

A simple chatbot without domain restriction was used as a baseline. This system generated responses without verifying data from the knowledge base.

Evaluation Metrics

The system was evaluated using the following metrics:

- Accuracy: Percentage of correct responses
- Relevance Score: How closely the response matches the query intent
- Error Rate: Percentage of incorrect or irrelevant responses

Testing Procedure

1. Input test queries into both systems (baseline and proposed)
2. Record responses
3. Compare outputs with expected answers
4. Calculate evaluation metrics

7. Sample Input and Output

Example 1: Product Inquiry

Input:

“What is the price of Toyota Corolla?”

Output:

“The price of Toyota Corolla is approximately 20 lakh. It is a reliable sedan suitable for daily use.”

Example 2: Product Recommendation

Input:

“Suggest a cheap car”

Output:

“You can consider Suzuki Alto, which is one of the most affordable options with good fuel efficiency.”

Example 3: Order Information

Input:

“How can I order a car?”

Output:

“You can place an order by contacting our sales team or visiting our website and submitting an order request form.”

Example 4: Out-of-Domain Query

Input:

“What is the weather today?”

Output:

“Sorry, I can only provide information related to our products and services.”

Limitations

Despite its effectiveness, the proposed system has several limitations:

- The chatbot is limited to the information available in the knowledge base and cannot handle queries outside the defined domain.
- The system relies on keyword-based retrieval, which may not capture complex user intent in some cases.
- The response generation is template-based and may lack deep conversational flexibility compared to advanced large language models.
- Scalability may become an issue if the knowledge base grows significantly without optimization.

Conclusion

This paper presented a domain-specific customer support chatbot that utilizes Retrieval-Augmented Generation to improve response accuracy and reliability. By integrating an external knowledge base and implementing a retrieval mechanism, the proposed system ensures that responses are based on verified business data.

The experimental results demonstrate that the system significantly reduces incorrect responses and enhances user satisfaction compared to traditional chatbot approaches. The domain restriction mechanism plays a crucial role in preventing irrelevant or misleading outputs, making the system suitable for real-world business applications.

Future Work

Future improvements can be made to enhance the system's capabilities:

- Integrating advanced semantic search techniques (e.g., vector embeddings) for better query understanding
- Incorporating user context and conversation memory for more personalized interactions
- Expanding the system to support multilingual communication
- Enhancing response generation using advanced language models for more natural conversations
- Deploying the system in real-world business environments for large-scale evaluation

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Declaration

I hereby declare that this project report and the accompanying implementation are entirely my own work, completed as part of the requirements for CSE 342 (Artificial Intelligence Sessional). The project has not been submitted to any other course or institution. All sources of information consulted have been duly acknowledged in the References section. The code included in this report has been written, tested, and understood by me personally.

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